ECE3411

Lecture9

# PWM: Pulse Width Modulation

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Copied from Lecture 3b, ECE3411 – Fall 2015, by
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Adapted from John Chandy ECE3411 – Fall 2024

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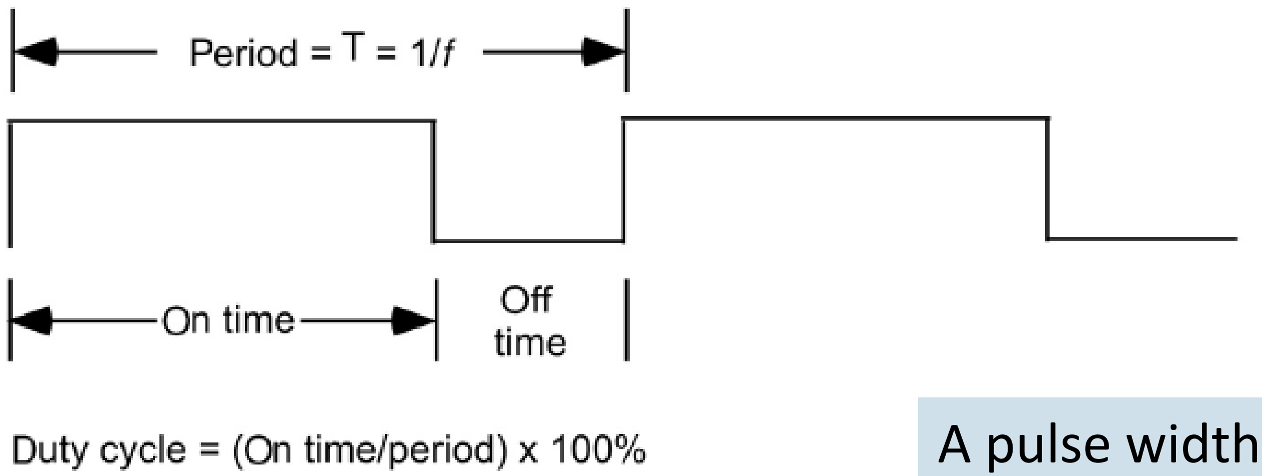
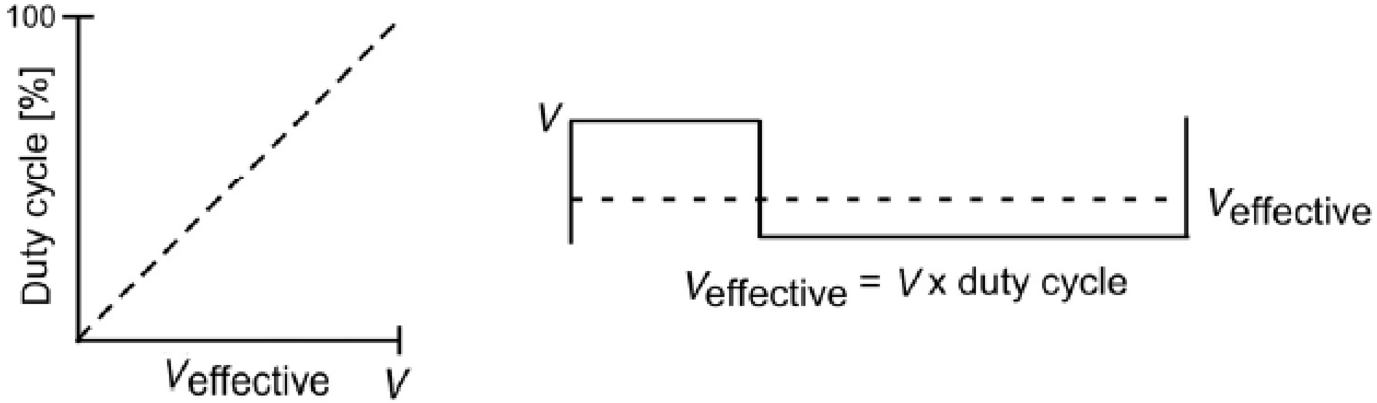
# Pulse Width Modulation



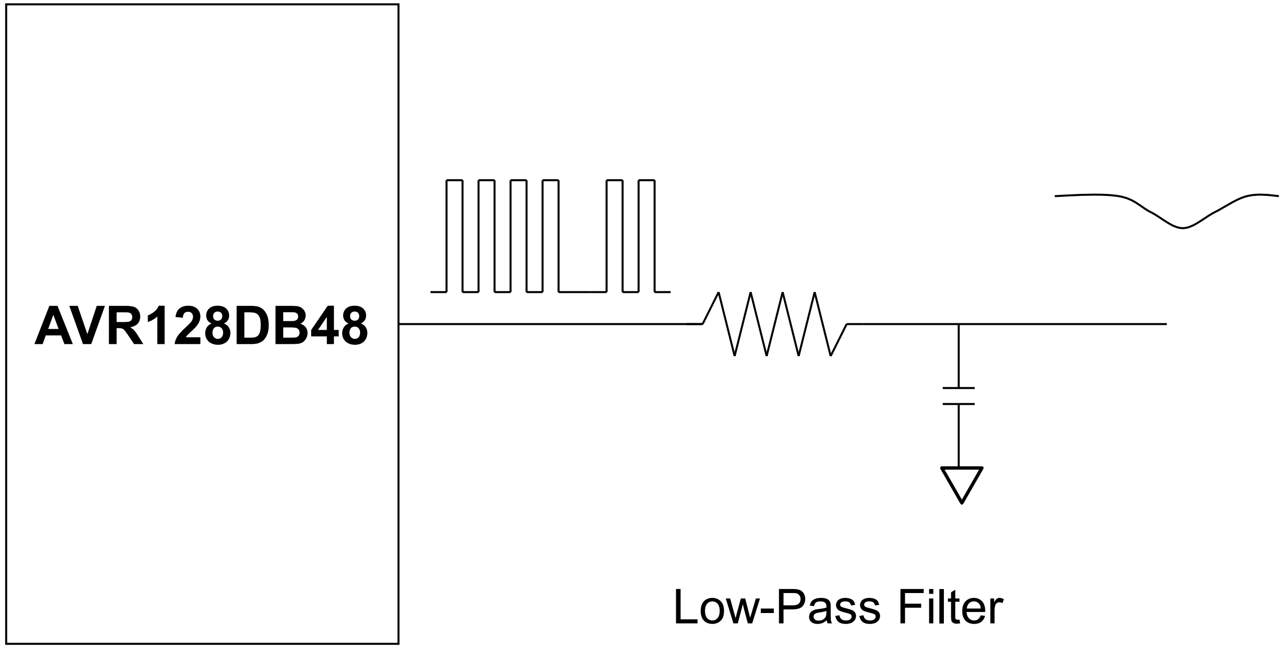

(a) Signal parameters

A pulse width modulated (PWM) signal is characterized by a fixed frequency and a varying duty cycle.




(b) Pulse width modulation (PWM) concepts

# Pulse Width Modulation



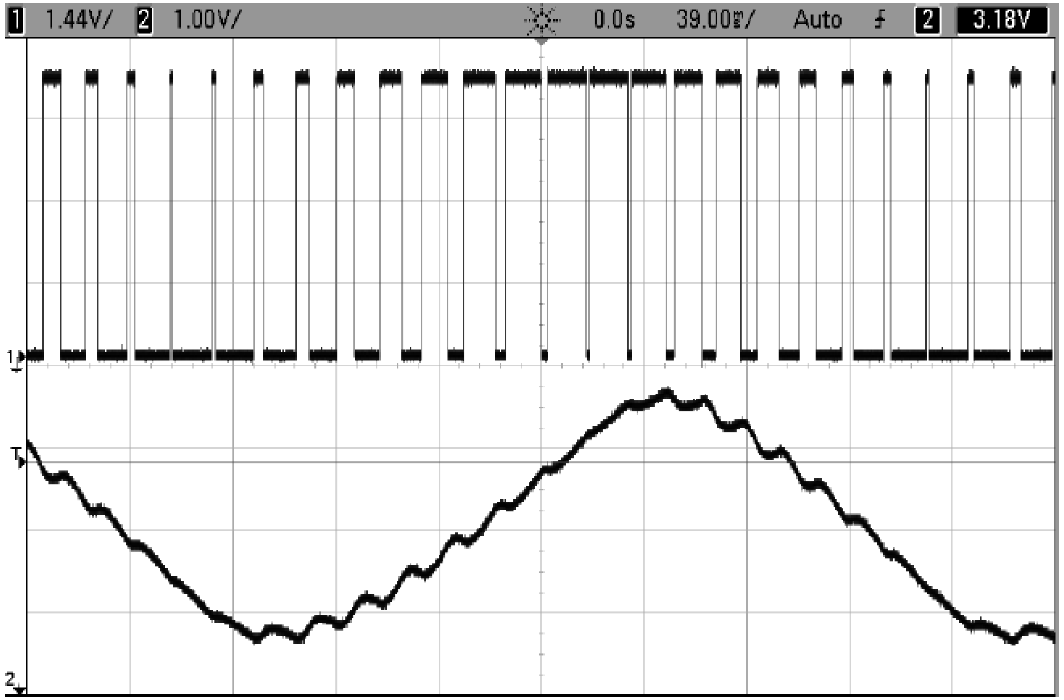
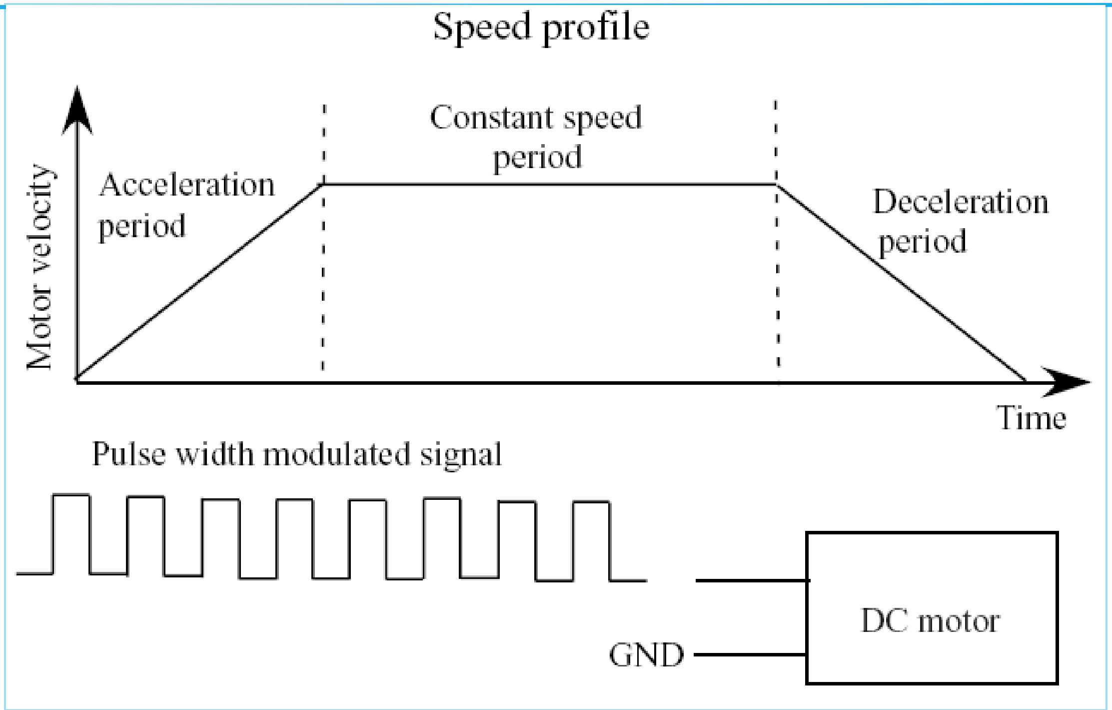
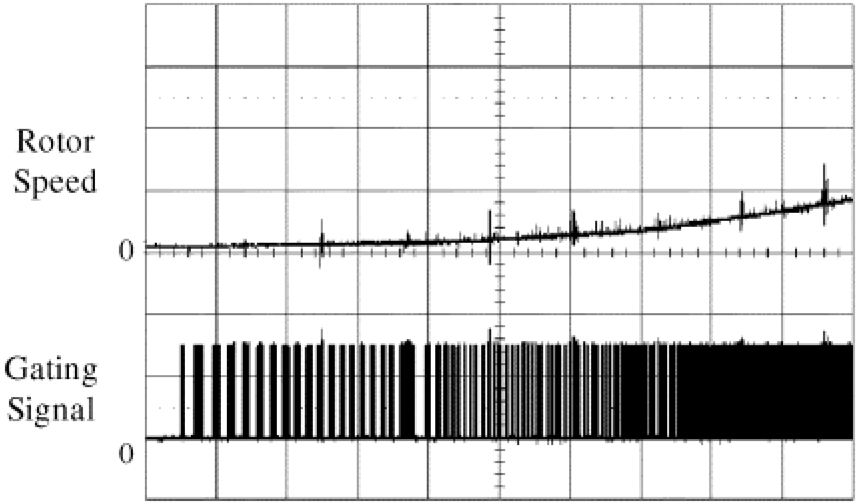
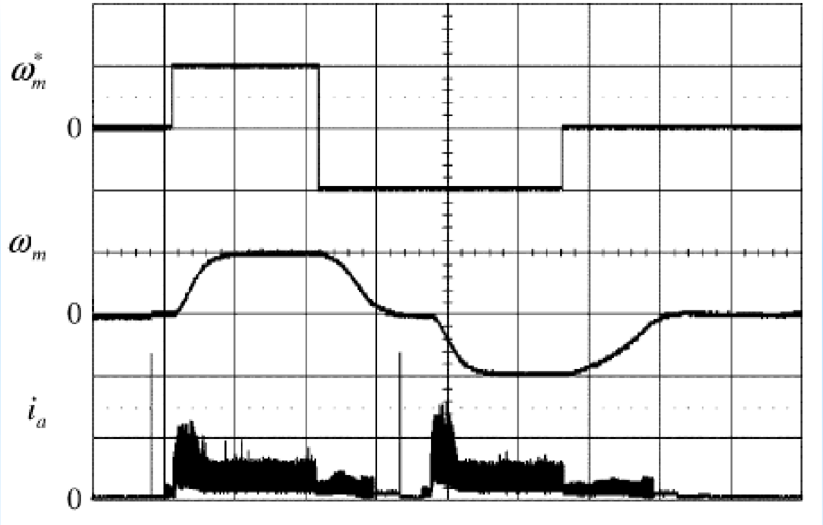




**Figure 10-1.** *PWM oscilloscope traces*

# PWM Application: Motor Control



# Pulse Width Modulation

- If duty cycle is *d*, the width of each pulse is *Td*, where *T* is the length of each period. *T* is the inverse of the switching frequency.
- Resolution of each pulse is determined by the $f_{CLK}$
- If you want 8-bit resolution, that means you need 256 possible pulse widths. Thus, the smallest *T* you can have is $\frac{256}{f_{CLK}}$.
- More generally, the highest possible switching frequency is $\frac{f_{CLK}}{2^n}$ for *n*–bit resolution

# Generating PWM with AVR128DB48

- Connect the PWM signal to one of your output pins
- How do you alter the square wave in code?
- It’s too fast  to do in software

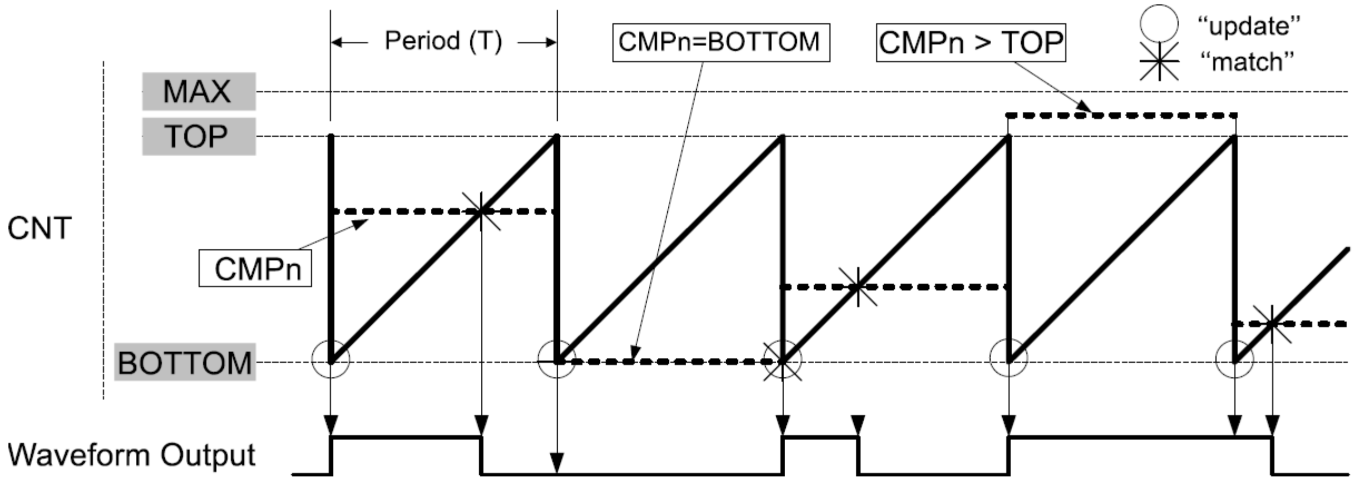
– 10kHz at 8-bit means the pulse width resolution is ~400ns.

- AVR timers can automatically generate the PWM signals

# Timer: PWM Modes

- PWM is generated with timers
- PER controls the PWM period/frequency
- CMPn controls the duty cycle

**Figure 23-12. Single-Slope Pulse-Width Modulation**

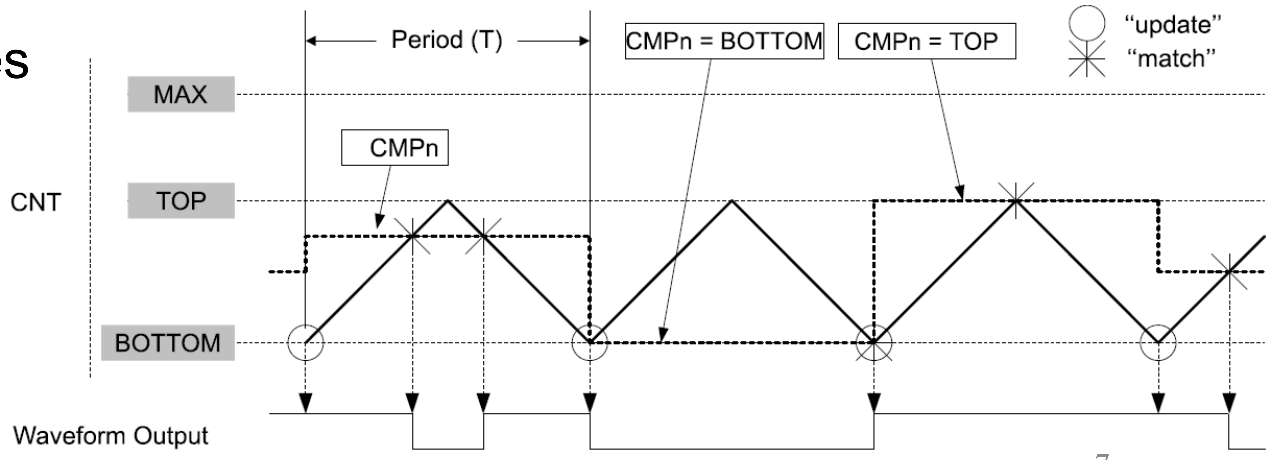



- Two modes:
- Single slop can have higher PWM frequencies

  (power converter applications)
- Dual slop is phase accurate

  (motor drive applications)

**Figure 23-13. Dual-Slope Pulse-Width Modulation**



# TC0A.SINGLE.CTRLB

### 21.5.2 Control B - Normal Mode

| Bit | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| | | CMP2EN | CMP1EN | CMP0EN | ALUPD | WGMODE[2:0] | | |
| Access | | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Reset | | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**Bits 4, 5, 6 – CMPEN** Compare n Enable
In the FRQ and PWM Waveform Generation modes, the Compare n Enable (CMPnEN) bits will make the waveform output available on the pin corresponding to WOn, overriding the value in the corresponding PORT output register. The corresponding pin direction must be configured as an output in the PORT peripheral.

| Value | Description |
|---|---|
| 0 | Waveform output WOn will not be available on the corresponding pin |
| 1 | Waveform output WOn will override the output value of the corresponding pin |

# TC0A.SINGLE.CTRLB

## 21.5.2 Control B - Normal Mode

| Bit | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| | | CMP2EN | CMP1EN | CMP0EN | ALUPD | WGMODE[2:0] | | |
| Access | | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Reset | | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

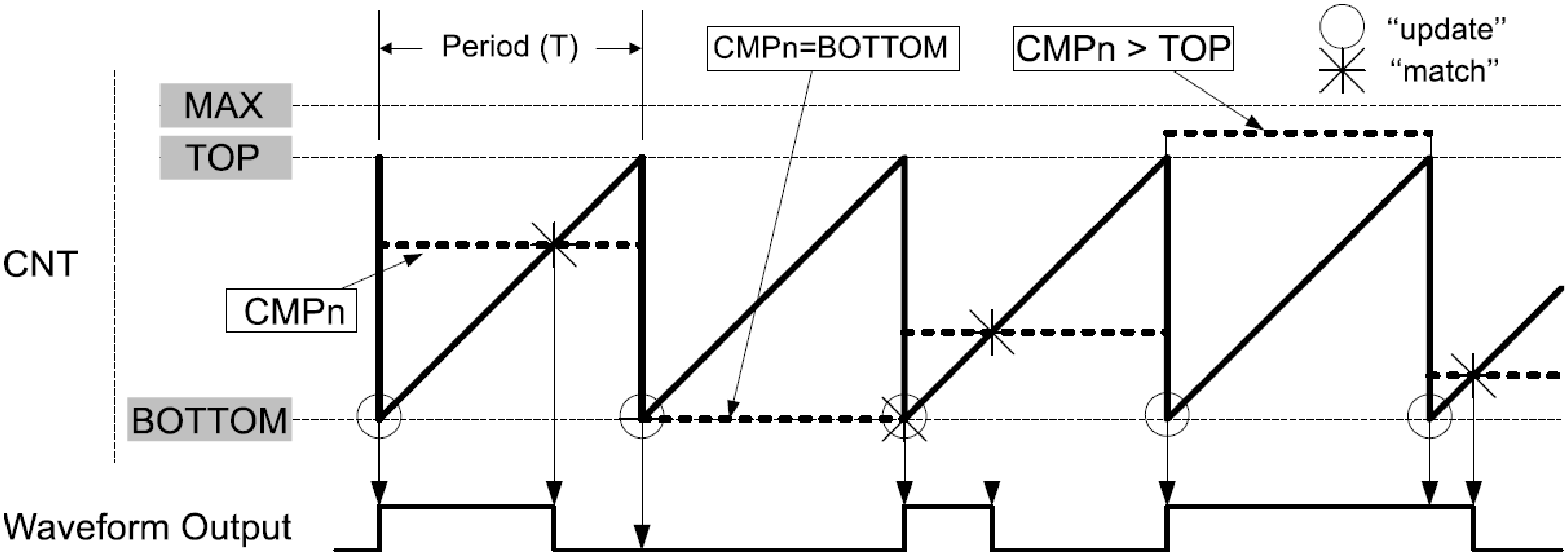
**Bits 2:0 – WGMODE[2:0]** Waveform Generation Mode
This bit field selects the Waveform Generation mode and controls the counting sequence of the counter, TOP value, UPDATE condition, Interrupt condition, and the type of waveform generated.
No waveform generation is performed in the Normal mode of operation. For all other modes, the waveform generator output will only be directed to the port pins if the corresponding CMPnEN bit has been set. The port pin direction must be set as output.

**Table 21-7. Timer Waveform Generation Mode**

| Value | Group Configuration | Mode of Operation | TOP | UPDATE | OVF |
|---|---|---|---|---|---|
| 0x0 | NORMAL | Normal | PER | TOP(1) | TOP(1) |
| 0x1 | FRQ | Frequency | CMP0 | TOP(1) | TOP(1) |
| 0x2 | - | Reserved | - | - | - |
| 0x3 | SINGLESLOPE | Single-slope PWM | PER | BOTTOM | BOTTOM |
| 0x4 | - | Reserved | - | - | - |
| 0x5 | DSTOP | Dual-slope PWM | PER | BOTTOM | TOP |
| 0x6 | DSBOTH | Dual-slope PWM | PER | BOTTOM | TOP and BOTTOM |
| 0x7 | DSBOTTOM | Dual-slope PWM | PER | BOTTOM | BOTTOM |

# Timer: PWM Modes

**Figure 23-12. Single-Slope Pulse-Width Modulation**



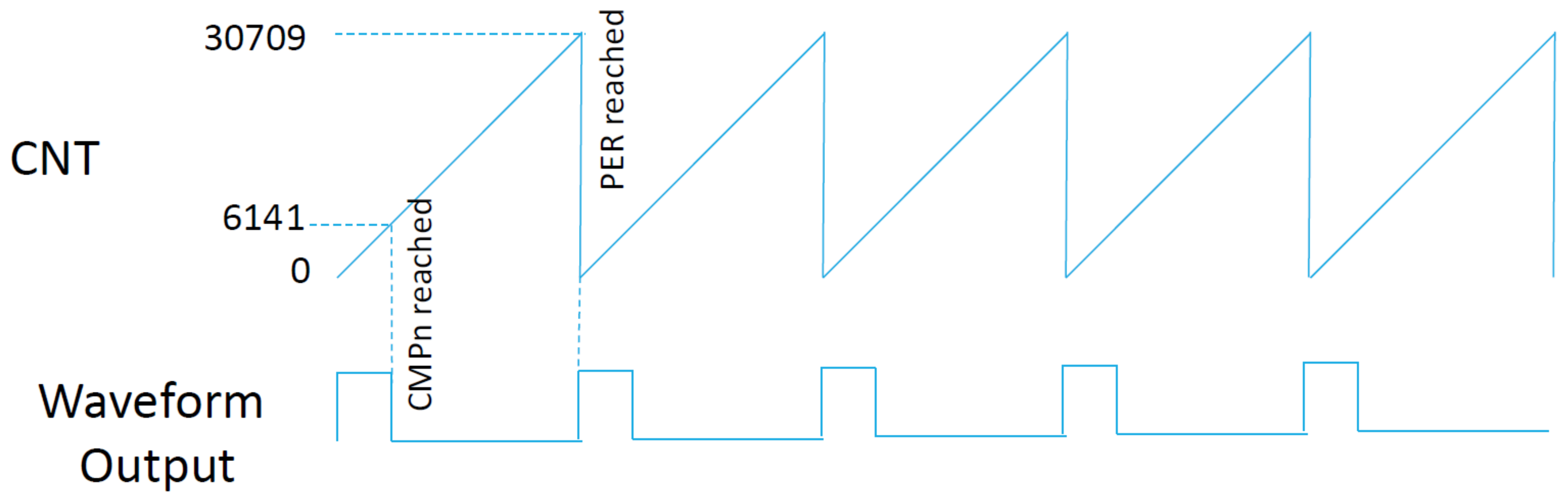
# Frequency & Duty Cycle

- PWM frequency: $\frac{f_{CLK\_PER}}{N(PER+1)}$

For example, if we want a PWM frequency of 521 Hz with F_CPU = 16 MHz and no prescalar, we need to set PER = 30709

- $DutyCycle = \frac{1+CMPn}{1+PER}$

If we want a 20% duty cycle with PER = 30709, and no prescalar, we get CMPn = 6141



# Example Calculations

- Assume a clock frequency of F_CPU=16 MHz.
- In single-slope PWM mode, configure a PWM output signal so that it is active high for 1/3 of a 12ms period:
- Which prescalar for the timer do you use? Given the prescalar of your choice, at what values should you set PER and CMPn?

- Prescalar of 1
- 1/16MHz = 62.5 ns / CNT tick.
- 12ms = 12ms/62.5ns CNT ticks, i.e., 192,000 CNT ticks.
- Too big for PER, so prescalar of 1 does not work

# Example Calculations

- Prescalar of 8
- 8/16MHz = 0.5μs/ CNT tick
- 12ms = 12ms/0.5us CNT ticks, i.e., 24,000 CNT ticks.
- By setting PER = 23999, we get a PWM of period 12 ms
- By setting CMPn = 7999, we get a duty cycle of (CMPn+1)/(PER+1) = 1/3.

- Prescalar of 1024
- 1024/16MHz = 64μs / CNT tick
- 12ms = 12ms/64us CNT ticks, i.e., 187.5 CNT ticks
- By setting PER = 188 we get a PWM of period 12.032 ms
- By setting CMPn = 63 we get a duty cycle of (CMPn+1)/(PER+1) slightly larger than 1/3.

# Waveform Generation

The compare channels can be used for waveform generation on the corresponding port pins. The following requirements must be met to make the waveform visible on the connected port pin:

1. A Waveform Generation mode must be selected by writing the Waveform Generation Mode (WGMODE) bit field in the TCAn.CTRLB register.

2. The used compare channels must be enabled (CMPnEN = 1 in TCAn.CTRLB), which will override the output value for the corresponding pin. An alternative pin can be selected by configuring the Port Multiplexer – Port Multiplexer section for details.

3. The direction for the associated port pin n must be configured in the Port peripheral as an output.

4. Optional: Enable the inverted waveform output for the associated port pin n. Refer to the section for details.

# Port Register Settings for PWM

- Besides registers PER and CMPn which other registers *need* to be set in order to have pin B1 output the PWM signal?
- Pin B1 is the CMP1 waveform output on TCA1

```
// pin PB1 = 1,WO1 is an output
PORTB.DIRSET = PIN1_bm;

// Single slope PWM mode
TCA1.SINGLE.CTRLB = TCA_SINGLE_WGMODE_SINGLESLOPE_gc;

// Enable CMP1 waveform output
TCA1.SINGLE.CTRLB |= TCA_SINGLE_CMP1EN_bm;

// Set Prescalar to 8 & enable timer.
TCA1.SINGLE.CTRLA |= (TCA_SINGLE_CLKSEL_DIV8_gc |
                      TCA_SINGLE_ENABLE_bm);
```

# TCAROUTEA: TCAn Pin Position

| Bit | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| | | | TCA1[2:0] | | | TCA0[2:0] | | |
| Access | | | R/W | R/W | R/W | R/W | R/W | R/W |
| Reset | | | 0 | 0 | 0 | 0 | 0 | 0 |

**Bits 5:3 – TCA1[2:0]** TCA1 Signals
This bit field controls the pin positions for TCA1 signals.

| Value | Name | Description | | | | | |
|---|---|---|---|---|---|---|---|
| | | WO0 | WO1 | WO2 | WO3 | WO4 | WO5 |
| 0x0 | PORTB | PB0 | PB1 | PB2 | PB3 | PB4 | PB5 |
| 0x1 | PORTC | PC4 | PC5 | PC6 | - | - | - |
| 0x2 | PORTE | PE4 | PE5 | PE6 | - | - | - |
| 0x3 | PORTG | PG0 | PG1 | PG2 | PG3 | PG4 | PG5 |
| Other | - | Reserved | | | | | |

**Bits 2:0 – TCA0[2:0]** TCA0 Signals
This bit field controls the pin positions for TCA0 signals.

| Value | Name | Description | | | | | |
|---|---|---|---|---|---|---|---|
| | | WO0 | WO1 | WO2 | WO3 | WO4 | WO5 |
| 0x0 | PORTA | PA0 | PA1 | PA2 | PA3 | PA4 | PA5 |
| 0x1 | PORTB | PB0 | PB1 | PB2 | PB3 | PB4 | PB5 |
| 0x2 | PORTC | PC0 | PC1 | PC2 | PC3 | PC4 | PC5 |
| 0x3 | PORTD | PD0 | PD1 | PD2 | PD3 | PD4 | PD5 |
| 0x4 | PORTE | PE0 | PE1 | PE2 | PE3 | PE4 | PE5 |
| 0x5 | PORTF | PF0 | PF1 | PF2 | PF3 | PF4 | PF5 |
| 0x6 | PORTG | PG0 | PG1 | PG2 | PG3 | PG4 | PG5 |
| 0x7 | - | Reserved | | | | | |

# PWM sample code

- Pin B2 is the CMP2 waveform output on TCA0

```
void TCA0_PWM_Init() {
// Enable PORTMUX to route TCA0 outputs to PORTB (PB0-PB5)
      PORTMUX.TCAROUTEA = PORTMUX_TCA0_PORTB_gc;

// Configure PB2 as an output
      PORTB.DIRSET = PIN2_bm;

// Enable TCA0 in Single-Slope PWM Mode
      TCA0.SINGLE.CTRLA = TCA_SINGLE_CLKSEL_DIV64_gc | TCA_SINGLE_ENABLE_bm;
// Enable TCA0 with prescaler 64
      TCA0.SINGLE.CTRLB = TCA_SINGLE_WGMODE_SINGLESLOPE_gc | TCA_SINGLE_CMP2EN_bm;
// Enable PWM on CMP2 (PB2)

// Set PWM Period and Duty Cycle
      TCA0.SINGLE.PER = 1000;   // Set PWM Period
      TCA0.SINGLE.CMP2BUF = 500;   // 50% Duty Cycle (PB2 toggles ON/OFF at 50%)
}
```

# Interrupts

### 23.5.10 Interrupt Control Register - Normal Mode

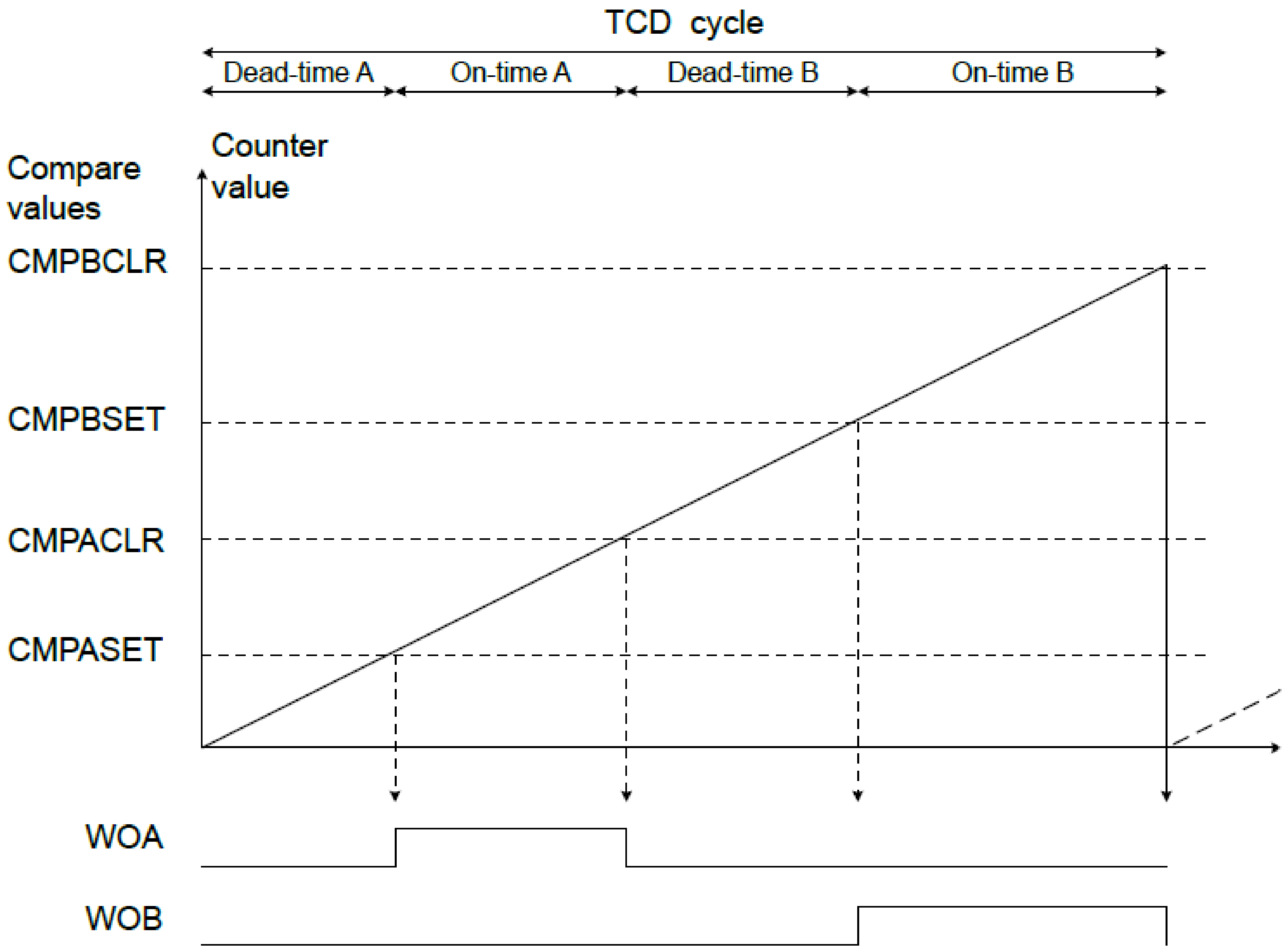
| Bit | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| | | CMP2 | CMP1 | CMP0 | | | | OVF |
| Access | | R/W | R/W | R/W | | | | R/W |
| Reset | | 0 | 0 | 0 | | | | 0 |

Allows one to change output on every PWM cycle

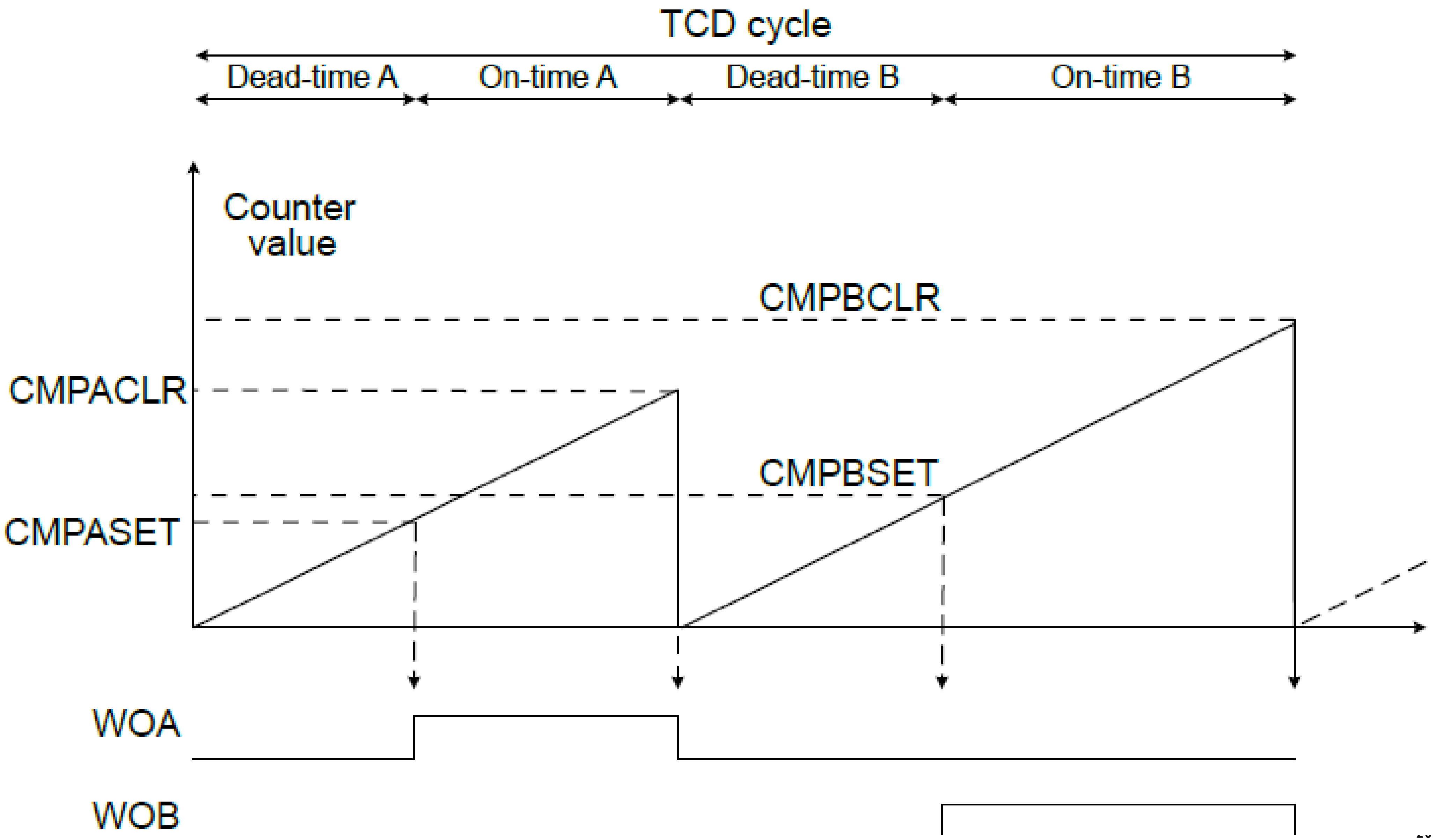
Modulate duty cycle in ISR (change CMPn)

- Remember that we don’t want to modify PER or CMPn to a value <CNT
  - Use double buffering: PERBUF or CMPnBUF

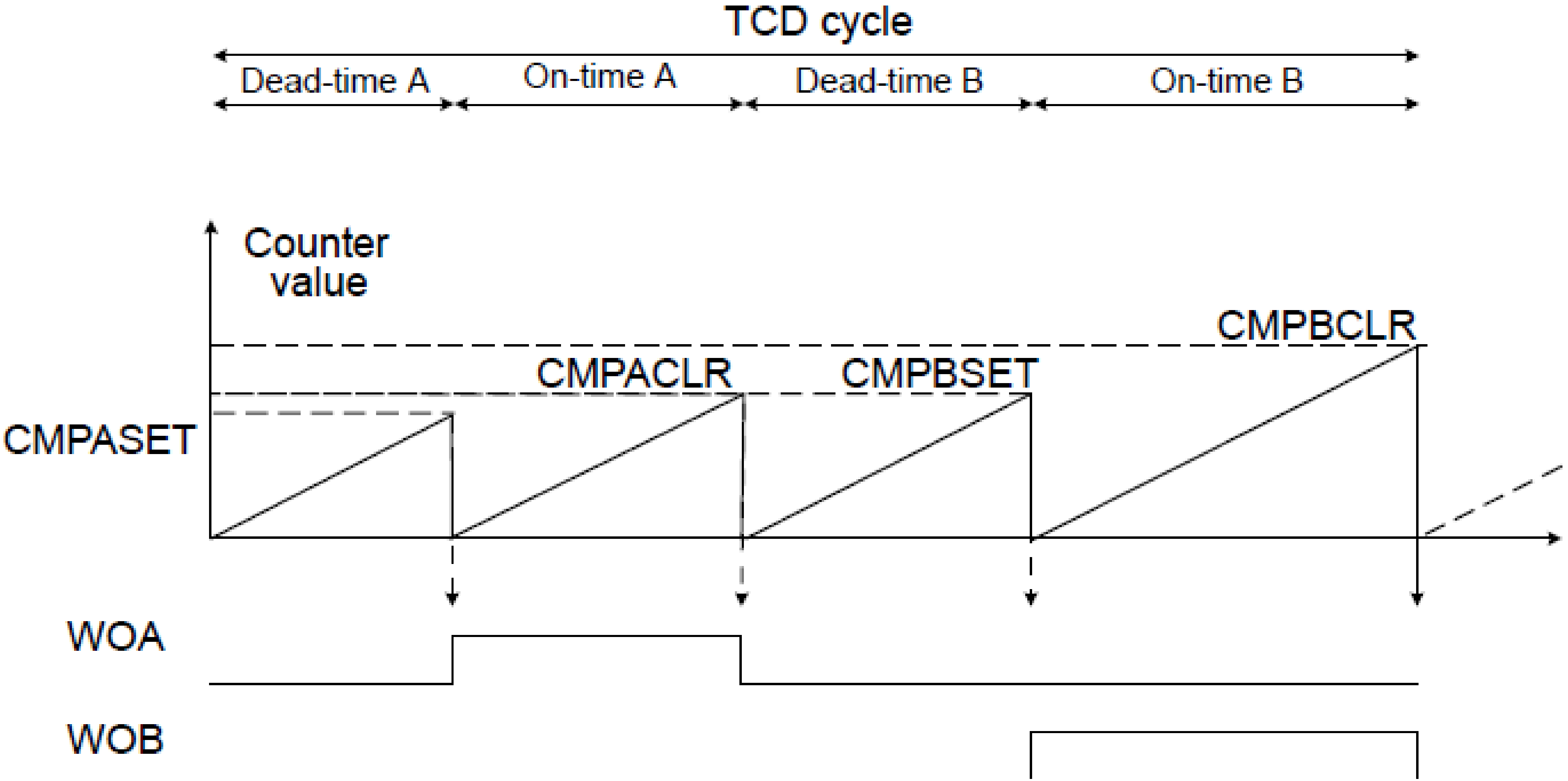
# One Ramp Mode



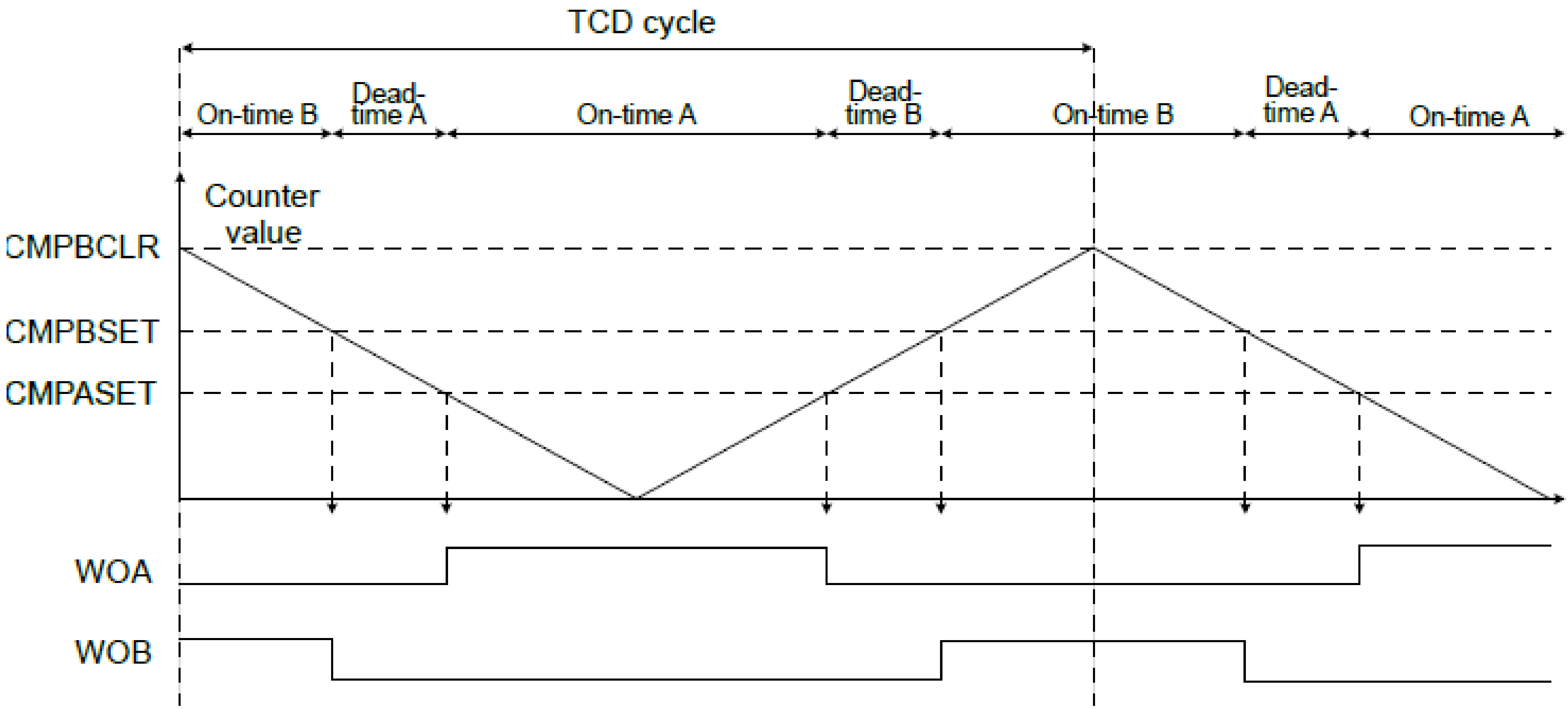
# Two Ramp Mode



# Four Ramp Mode



# Dual Slop Mode with Dead time



# Lab practice 8: LED Brightness Control

Use a 16-bit Timer/Counter Type A to generate a single slope PWM signal on PORTD such that:

- The PWM signal has a frequency of 1000Hz
- While button switch 1 is pressed, the duty cycle of the PWM signal is gradually increased (in steps of 1%) up to 95%
- While button switch 2 is pressed, the duty cycle of the PWM signal is gradually decreased (in steps of 1%) down to 5%